

Dell EMC NetWorker

Version 9.2

REST API Getting Started Guide

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REV 01

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Preface

As part of an effort to improve its product lines, EMC periodically releases revisions of its software and hardware. Therefore, some functions that are described in this document might not be supported by all versions of the software or hardware currently in use. The product release notes provide the most up-to-date information on product features.

Contact your EMC technical support professional if a product does not function correctly or does not function as described in this document.

Note

This document was accurate at publication time. Go to EMC Online Support (<https://support.emc.com>) to ensure that you are using the latest version of this document.

Purpose

This document describes how to configure and use NetWorker REST API.

Audience

This guide is part of the NetWorker documentation set, and is intended for use by developers who are creating programmatic interfaces to NetWorker systems.

Revision history

The following table presents the revision history of this document.

Table 1 Revision history

Revision	Date	Description
01	June, 2017	First release of this document for NetWorker 9.2.

Related documentation

The NetWorker documentation set includes the following publications, available on the Support website:

- *EMC NetWorker Online Software Compatibility Guide*
Provides compatibility information, including specific software and hardware configurations that NetWorker supports. To access the matrix, go to <http://compatibilityguide.emc.com:8080/CompGuideApp/>.
- *EMC NetWorker Administration Guide*
Describes how to configure and maintain the NetWorker software.
- *EMC NetWorker Network Data Management Protocol (NDMP) User Guide*
Describes how to use the NetWorker software to provide data protection for NDMP filers.
- *EMC NetWorker Cluster Integration Guide*
Contains information related to configuring NetWorker software on cluster servers and clients.
- *EMC NetWorker Installation Guide*
Provides information on how to install, uninstall, and update the NetWorker software for clients, storage nodes, and servers on all supported operating systems.

- *EMC NetWorker Updating from a Previous Release Guide*
Describes how to update the NetWorker software from a previously installed release.
- *EMC NetWorker Release Notes*
Contains information on new features and changes, fixed problems, known limitations, environment and system requirements for the latest NetWorker software release.
- *EMC NetWorker Command Reference Guide*
Provides reference information for NetWorker commands and options.
- *EMC NetWorker Data Domain Boost Integration Guide*
Provides planning and configuration information on the use of Data Domain devices for data deduplication backup and storage in a NetWorker environment.
- *EMC NetWorker Performance Optimization Planning Guide*
Contains basic performance tuning information for NetWorker.
- *EMC NetWorker Server Disaster Recovery and Availability Best Practices Guide*
Describes how to design, plan for, and perform a step-by-step NetWorker disaster recovery.
- *EMC NetWorker Snapshot Management Integration Guide*
Describes the ability to catalog and manage snapshot copies of production data that are created by using mirror technologies on storage arrays.
- *EMC NetWorker Snapshot Management for NAS Devices Integration Guide*
Describes how to catalog and manage snapshot copies of production data that are created by using replication technologies on NAS devices.
- *EMC NetWorker Security Configuration Guide*
Provides an overview of security configuration settings available in NetWorker, secure deployment, and physical security controls needed to ensure the secure operation of the product.
- *EMC NetWorker VMware Integration Guide*
Provides planning and configuration information on the use of VMware in a NetWorker environment.
- *EMC NetWorker Error Message Guide*
Provides information on common NetWorker error messages.
- *EMC NetWorker Licensing Guide*
Provides information about licensing NetWorker products and features.
- *EMC NetWorker REST API Getting Started Guide*
Describes how to configure and use the NetWorker REST API to create programmatic interfaces to the NetWorker server.
- *EMC NetWorker REST API Reference Guide*
Provides the NetWorker REST API specification used to create programmatic interfaces to the NetWorker server.
- Describes the integration of NetWorker with CloudBoost.
- EMC NetWorker Management Console Online Help
Describes the day-to-day administration tasks performed in the NetWorker Management Console and the NetWorker Administration window. To view the online help, click **Help** in the main menu.
- EMC NetWorker User Online Help
Describes how to use the NetWorker User program, which is the Windows client interface, to connect to a NetWorker server to back up, recover, archive, and retrieve files over a network.

Special notice conventions that are used in this document

EMC uses the following conventions for special notices:

NOTICE

Identifies content that warns of potential business or data loss.

Note

Contains information that is incidental, but not essential, to the topic.

Typographical conventions

EMC uses the following type style conventions in this document:

Table 2 Style conventions

Bold	Used for names of interface elements, such as names of buttons, fields, tab names, and menu paths (what the user specifically selects or clicks)
<i>Italic</i>	Used for full titles of publications that are referenced in text
Monospace	Used for: • System code • System output, such as an error message or script • Pathnames, file names, prompts, and syntax • Commands and options
<i>Monospace italic</i>	Used for variables
Monospace bold	Used for user input
[]	Square brackets enclose optional values
	Vertical bar indicates alternate selections - the bar means “or”
{ }	Braces enclose content that the user must specify, such as x or y or z
...	Ellipses indicate non-essential information that is omitted from the example

Where to get help

EMC support, product, and licensing information can be obtained as follows:

Product information

For documentation, release notes, software updates, or information about EMC products, go to EMC Online Support at <https://support.emc.com>.

Technical support

Go to EMC Online Support at <https://support.emc.com>, and click **Service Center**. Several options for contacting EMC Technical Support appear on the site. Note that to open a service request, you must have a valid support agreement. Contact your EMC sales representative for details about obtaining a valid support agreement or with questions about your account.

Online communities

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engage online with customers, partners, and certified professionals for all EMC products.

Your comments

Your suggestions help to improve the accuracy, organization, and overall quality of the user publications. Send your opinions of this document to

DPAD.Doc.Feedback@emc.com.

CHAPTER 1

Introduction

This chapter includes the following topics:

- [Overview of the NetWorker REST API](#) 12
- [Installation](#) 12
- [Getting started with the NetWorker REST API](#) 12
- [Authentication and authorization](#) 14
- [New features in the NetWorker REST API for release 9.2](#) 15

Overview of the NetWorker REST API

The NetWorker REST API is an interface that provides programmatic access to the NetWorker data protection service. By using this REST API, customers can build client applications to automate NetWorker operations. This document describes how to access the NetWorker data protection service by using the NetWorker REST API, and provides example tutorials for common tasks.

NetWorker REST API service

The NetWorker REST API uses a service to facilitate operations. The NetWorker REST API service is deployed in the same Apache tomcat container as the NetWorker authentication service. The NetWorker REST API uses the same set of tomcat processes to deliver its service.

API versions for the NetWorker REST API

There are three versions of the NetWorker REST API, which correspond to releases of the NetWorker software. Versions of the REST API can be viewed at the root URI of `https://{nw-server-hostname}:9090/nwrestapi`.

API version	NetWorker release	URI
NetWorker REST API v1	NetWorker 9.0 and minor releases	<code>https://nw-server-hostname:9090/nwrestapi/v1</code>
NetWorker REST API v2	NetWorker 9.1 and minor releases	<code>https://nw-server-hostname:9090/nwrestapi/v2</code>
NetWorker REST API v3	NetWorker 9.2	<code>https://nw-server-hostname:9090/nwrestapi/v3</code>

Code written using a specific version of the API will continue to function when subsequent versions of the API are released. However, in some cases, older versions of certain resources might be deprecated in the future. NetWorker only deprecates resources when necessary. If an older version of a resource can continue to function, it is accessible at its original API version number URI.

Installation

The NetWorker REST API is installed as part of NetWorker installation.

As part of the installation process, an Apache Tomcat instance is installed and a non-root user, `nsrtomcat`, is created. If your system has special user security requirements, ensure that proper operational permissions are granted to the `nsrtomcat` users.

The *EMC NetWorker Installation Guide* contains further information.

Getting started with the NetWorker REST API

This document provides example tutorials about how to perform some of the common tasks that programmers working with the NetWorker software will want to accomplish. After working through these tutorials, the use of the API should be

familiar enough that you will be able to develop interfaces that make use of the full range of available functionality.

NetWorker REST API specification

The complete specification for the NetWorker REST API is available in PDF format in the *EMC NetWorker REST API Reference Guide*.

Note

The NetWorker REST API is targeted to automate daily backup tasks. One-time configuration tasks, such as client and device configuration, should be done outside of the REST API using NetWorker Management Console wizards.

Using the example tutorials in this guide

The example tutorials provided in this guide follow a similar pattern, first presenting an operation that achieves a particular objective (for example, creating a client) in the form of a NetWorker command line operation that can be run locally. Next, the same objective is presented in the form of a NetWorker REST API call. Users familiar with the NetWorker command line will recognize the command line form of these operations.

Because of limited command line support for VMware client operations, the examples in [Working with VMware-based clients](#) on page 29 use NetWorker Management Console examples, rather than command line examples.

This document uses an example NetWorker server installed on a server named mars, and NetWorker REST API is exposed in the following URL: `https://mars:9090/nwrestapi/`.

When following these tutorials, you can use a curl client, or any other HTTP client. Typical curl syntax to call the API will look as follows:

```
curl -X POST -H "Content-Type: application/json" -H "Authorization: Basic YWRtaW5pc3RyYXRvcjohNFUyYnVpbGQ=" -d '{ "hostname" : "saturn" }' "https://mars:9090/nwrestapi/v2/global/clients" -k -l
```

The following table examines the parameters being passed into the curl command in the above example.

Parameter	Value
-X POST	This specifies the HTTP method that is being used. Generally: - The <code>GET</code> method is used to retrieve data. - The <code>PUT</code> method is used to update data. - The <code>POST</code> method is used to create. - The <code>DELETE</code> method is used to remove.
-H "Content-Type: application/json"	Specifies the type of payload being sent to the API. The NetWorker REST API uses json exclusively, so this never changes.

Parameter	Value
-H "Authorization: Basic YWRtaW5pc3RyYXRvcjohNFUyYnVpbGQ="	Specifies an authorization header that contains Base64 encoded username and password.
-d '{ "hostname" : "saturn" }'	Specifies the actual payload being sent to the API.
https://mars:9090/nwrestapi/v2/global/ clients	Specifies the URL of the API endpoint to which the request is being sent.
-k -1	Specifies version of the SSL protocol to be used for communication with the API. This option is environment specific.

To make the examples more readable, examples in this document simplify the API request into tool independent form, where only the data that actually changes between requests is specified (generally the HTTP method, endpoint URL, and the actual payload). For the endpoint URL, the hostname in the URL will also be ignored (for example, https://mars:9090).

So the example will in fact be denoted as:

```
POST /global/clients{ "hostname" : "saturn" }
```

Note that the payload is only required for PUT and POST requests.

Authentication and authorization

With the NetWorker REST API, all endpoints other than initial landing endpoint (<https://your-server-name:9090/nwrestapi/>) require authentication.

The NetWorker REST API uses the standard, basic HTTP authentication as described https://en.wikipedia.org/wiki/Basic_access_authentication. Basic HTTP Authentication is a common standard and is supported by most common HTTP tools such as curl, Postman, web browsers and programming frameworks.

When you are using Basic HTTP Authentication, user credentials are passed to the API with every request by using the authorization header. The authorization header takes the form of "Basic *token*," where the *token* is a Base64-encoded string in the format username:password.

It is important to note that Base64 encoding is not an encryption, and does not provide any protection for the password being used. Therefore, it is essential for security purposes that the NetWorker REST API uses HTTPS, which provides transport layer encryption for all API interactions.

Once a user has been authenticated by the API, permissions to NetWorker resources will be based on the user's NetWorker permissions.

New features in the NetWorker REST API for release 9.2

This section summaries new features and changes in the NetWorker REST API for the 9.2 release of NetWorker. The API changes for NetWorker 9.2 are in v3 of the REST API, found at the URI `https://nw-server-hostname:9090/nwrestapi/v3`.

There have been no changes to versions 1 and 2 of the NetWorker REST API in the 9.2 release of NetWorker. You can continue to use code written for those versions of the REST API without making any changes to the code.

The *NetWorker REST API Reference Guide* has further information about the new parameters discussed in this section.

Support for Data Domain retention lock

Support for Data Domain retention lock has been added to v3 of the NetWorker REST API.

- The following new parameters have been added to the Device resource:
 - `dataDomainRetentionLockMode`
 - `dataDomainRetentionLockPeriodMin`
 - `dataDomainRetentionLockPeriodMax`
- The following new parameters have been added to the PolicyActionBackupTraditional resource:
 - `enableDDRetentionLock`
 - `ddRetentionLockTime`
- The following new parameters have been added to the PolicyActionClone resource:
 - `enableDDRetentionLock`
 - `ddRetentionLockTime`

Change of URI for tenants

- The URI for tenants has been changed from `{base uri}/tenants` to `{base uri}/global/tenants`.
- URI for specific tenant is `{base uri}/global/tenants/{tenantId}`

Change of URI for usergroups

- URI for usergroups is `{base uri}/global/usergroups`.
- URI for specific tenant is `{base uri}/global/tenants/{tenantId}`

New parameter for Client resource

A new parameter, `storageNodes`, has been added to the Client resource:

New parameters for the PolicyActionBackupVMwareVProxy resource

The following new parameters have been added to the PolicyActionBackupVMwareVProxy resource:

- `appConsistentQuiesce`
- `transactionLogBackup`
- `quiesceTimeoutInMinutes`
- `quiesceSystemAdminUsername`

- `quiesceSystemAdminPassword`

Upgrading the NetWorker REST API

NetWorker release 9.2 includes support for version three (v3) of the NetWorker REST API. NetWorker release 9.2 includes support for v1 and v2 of the REST API, for backwards compatibility. All new features included in the REST API for NetWorker release 9.2 are in v3 of the API.

If you do not need any new functionality provided in v3 of the API, you can continue to use v1 or v2 endpoints and your existing application will work without requiring any changes. However, upgrading to v3 of the API is recommended. To do this you, will need to perform the following:

1. Regenerate your classes using either the Open API or JSON Schema definitions provided with v3 of the API.
2. The URIs for certain resources have changed in v3. Change the URIs in your code as necessary to accommodate these changes. [New features in the NetWorker REST API for release 9.2](#) on page 15 provides information about changed URIs. Also change the version numbers passed in the API URL from v1/v2 to v3.

CHAPTER 2

NetWorker REST API example usage

This chapter contains the following topics:

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• Searching for clients	21
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• Working with VMware-based clients	29

Creating a client

This section demonstrates how to create a client by using the NetWorker REST API.

Note

When creating module clients, use the Management Console Client Backup Configuration Wizard rather than the REST API.

Command line example

The following example command uses the `nsradmin` program to create a client:

```
nsradmin> create type: NSR client; name: saturn; save set: /etc/
hosts
                                type: NSR client;
                                name: saturn;
                                save set: /etc/hosts;
Create? y created resource id
46.0.236.47.0.0.0.0.71.53.50.87.10.5.167.140(1)
```

API request

The same client is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/clients
{
  "hostname" : "saturn",
  "saveSets" : ["/etc/hosts"]
}
```

API response

The following response is received from the API:

```
201 Created
```

Viewing all clients

This section demonstrates how to view all clients configured on a NetWorker server using the NetWorker REST API.

Command line example

The following example command uses the `nsradmin` program to list all client configured on the NetWorker server, and displays the returned data (in this case for one client, with the hostname "saturn"):

```
nsradmin> p type: NSR client
...
                                type: NSR client;
                                name: saturn;
                                comment: ;
                                directive: ;
                                protection group list: ;
                                save set: /etc/hosts;
                                Checkpoint enabled: Disabled;
```

```

Parallel save streams per save set: Disabled;
    remote access: ;
    remote user: ;
    password: ;
    NAS management user: ;
    NAS management password: ;
    NAS file access user: ;
    NAS file access password: ;
    index backup content: No;
    backup command: ;
    Pre command: ;
    Post command: ;
    application information: ;
    ndmp vendor information: ;
    ndmp: No;
    ndmp multi-streams enabled: No;
    Disable IPv6: No;
    NAS device: No;
    NDMP array name: ;
    NAS device management name: ;
    storage replication policy name: ;
    Probe resource name: ;
    Block based backup: No;
    executable path: ;
    aliases: saturn, saturn.company.com;
    parallelism: 12;
    backup type: ;
    tag: ;
    restricted data zone: ;

```

API request

The following command uses the NetWorker REST API to list all configured clients:

```
GET nwrestapi/v2/global/clients
```

API response

The following response is received from the API:

```

200 OK
{
  "clients": [
    ...
    {
      "aliases": [
        "saturn",
        "saturn.company.com"
      ],
      "applicationInformation": [],
      "blockBasedBackup": false,
      "checkpointEnabled": false,
      "clientId":
"5c088865-00000004-57339887-57339886-00010c00-80c64a29",
      "hostname": "saturn",
      "indexBackupContent": false,
      "links": [
        {
          "href": "https://mars:9090/nwrestapi/v2/global/clients/
48.0.254.77.0.0.0.0.149.148.51.87.128.222.109.201",
          "rel": "item"
        }
      ],
      "nasDevice": false,
      "ndmp": false,
    }
  ]
}

```

```

        "ndmpMultiStreamsEnabled": false,
        "ndmpVendorInformation": [],
        "parallelSaveStreamsPerSaveSet": false,
        "parallelism": 4,
        "protectionGroups": [],
        "remoteAccessUsers": [],
        "resourceId": {
            "id": "48.0.254.77.0.0.0.0.149.148.51.87.128.222.109.201",
            "sequence": 2
        },
        "saveSets": [
            "/etc/hosts"
        ],
        "scheduledBackup": true,
        "tags": []
    }
],
"count": 2
}

```

Viewing only selected fields for all clients

Because the example above listed all fields for all clients, the returned JSON objects will sometimes be very large. You can limit the number of fields that are returned by the NetWorker REST API, as follows.

Command line example

The following example command uses the `nsradmin` program to view all clients configured on a NetWorker, but restricting the returned data to only the client name and aliases (two clients are found in this example, with the hostnames "saturn" and "jupiter"):

```

nsradmin> show aliases; name
nsradmin> p type: NSR client
                name: saturn;
                aliases: saturn, saturn.company.com;

                name: jupiter;
                aliases: jupiter;

```

API request

The following command performs the same functionality using the NetWorker REST API:

```
GET nwrestapi/v2/global/clients?fl=aliases,hostname
```

API response

The following response is received from the API:

```

200 OK
{
  "clients": [
    {
      "aliases": [
        "jupiter"
      ],
      "hostname": "jupiter"
    }
  ]
}

```

```

    },
    {
      "aliases": [
        "saturn",
        "saturn.company.com"
      ],
      "hostname": "saturn"
    }
  ],
  "count": 2
}

```

Searching for clients

You can also restrict the number of clients that are found by using search criteria. In this case example, only one client with the hostname of "saturn" will be returned.

Command line example

The following example command uses the `nsradmin` program to search for a client:

```

nsradmin> . type: NSR client; name: saturn
Current query set
nsradmin>          name: saturn;
                  aliases: saturn, saturn.company.com;

```

API request

The same search is performed in the following example by using the NetWorker REST API:

```

GET /nwrestapi/v2/global/clients?
fl=aliases,hostname&q=hostname:saturn

```

API response

The following response is received from the API:

```

200 OK
{
  "clients": [
    {
      "aliases": [
        "saturn",
        "saturn.company.com"
      ],
      "hostname": "saturn"
    }
  ],
  "count": 1
}

```

Creating an Advanced File Type Device

This section demonstrates how to create an Advanced File Type Device (AFTD) using the NetWorker REST API.

Command line example

The following example command uses the `nsradmin` program to create an AFTD:

```
nsradmin> create type: NSR device; name: /space/storage; device
access information: /space/storage; media type: adv_file
                type: NSR device;
                name: /space/storage;
        device access information: /space/storage;
                media type: adv_file;
Create? y created resource id
47.0.236.47.0.0.0.0.71.53.50.87.10.5.167.140(1) nsradmin> q
```

API request

The same device is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/devices
{
  "name": "/space/storage",
  "deviceAccessInfo": "/space/storage",
  "mediaType": "adv_file"
}
```

API response

The following response is received from the API:

```
201 Created
```

Labeling a device

This section demonstrates how to label a device using the NetWorker REST API.

Command line example

The following example command uses the `nsrmm` command to label a device:

```
# nsrmm -l -m -y -b Default -f /space/storage
Using volume name `mars.001' for pool `Default'
```

API request

The device is labeled in the same manner in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/devices
{
  "name": "/space/storage",
  "deviceAccessInfo": "/space/storage",
```

```
"mediaType": "adv_file"
}
```

API response

The following response is received from the API:

```
202 Accepted
```

Creating a protection group

This section demonstrates how to create a protection group by using the NetWorker REST API.

Command line example

The following example command uses the `nsrpolicy` command to create a protection group :

```
# nsrpolicy group create client -g EngineeringWorkstations -C saturn
nsrpolicy: Group 'EngineeringWorkstations' was successfully created
```

API request

The same protection group is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/protectiongroups
{
  "workItemType" : "Client",
  "name" : "EngineeringWorkstations",
  "workItems" :
  ["48.0.254.77.0.0.0.0.149.148.51.87.128.222.109.201"]
}
```

Note

In the command line example, the name of the client is used. However when using the REST API, the `resourceId` of the client must be used instead, as show in this example in the `workItems` element. This can be retrieved from the `resourceId` element in the client resource, as shown in the API response section of [Viewing all clients](#) on page 18.

API response

The following response is received from the API:

```
201 Created
```

Creating a protection policy

This section demonstrates how to create a protection policy by using the NetWorker REST API.

Command line example

The following example command uses the `nsrpolicy` command to create a protection policy:

```
# nsrpolicy policy create -p Engineering
nsrpolicy: Policy 'Engineering' was successfully created
```

API request

The same protection policy is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/protectionpolicies
{
  "name": "Engineering"
}
```

API response

The following response is received from the API:

```
201 Created
```

Creating a workflow

This section demonstrates how to create a workflow by using the NetWorker REST API.

Command line example

The following example command uses the `nsrpolicy` command to create a workflow:

```
# nsrpolicy workflow create -p Engineering -w ClientProtection -g
EngineeringWorkstations -S 21:00
nsrpolicy: workflow 'ClientProtection' was successfully created
```

API request

The same workflow is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/protectionpolicies/Engineering/workflows
{
  "name" : "ClientProtection",
  "protectionGroups" : ["EngineeringWorkstations"],
  "startTime" : "21:00"
}
```

API response

The following response is received from the API:

```
201 Created
```

Creating actions for a traditional backup

This section demonstrates how to create actions for a traditional backup by using the NetWorker REST API.

Command line example

The following example command uses the `nsrpolicy` command to create a actions for a traditional backup:

```
# nsrpolicy action create backup traditional -p Engineering -w
ClientProtection -A Backup
nsrpolicy: Assigned default schedule period of 'week' and schedule
activity of 'full, incr, incr, incr, incr, incr, incr, incr'.
nsrpolicy: Action 'Backup' was given '1 Months' retention period
nsrpolicy: Action 'Backup' was given 'nsrserverhost' storage node
nsrpolicy: Action 'Backup' was given 'Default' destination pool
nsrpolicy: Action 'Backup' was successfully created
```

API request

The same actions are created in the following example by using the NetWorker REST API:

```
PUT /nwrestapi/v2/global/protectionpolicies/Engineering/workflows/
ClientProtection
{
  "actions":[
    {
      "name":"Backup",
      "actionSpecificData":{
        "backup":{
          "backupSpecificData":{
            "traditional":{}
          }
        }
      }
    }
  ]
}
```

API response

The following response is received from the API:

```
204 No Content
```

Performing an on-demand backup

This section demonstrates how to perform an on-demand backup of a client by using the NetWorker REST API.

Command line example

The following example commands uses the `nsrpolicy` command or the `nsrworkflow` command to perform an on-demand backup of a client:

```
# nsrpolicy start -p Engineering -w ClientProtection -c saturn:/etc/hosts
144091:nsrpolicy: Workflow 'Engineering/ClientProtection' started
and has job id 32023
```

or

```
# nsrworkflow -p Engineering -w ClientProtection -c saturn:/etc/hosts
133550:nsrworkflow: Starting Protection Policy 'Engineering'
workflow 'ClientProtection'.
123316:nsrworkflow: Starting action 'Engineering/ClientProtection/Backup'
with command: 'savegrp -Z backup:traditional -v'.
123321:nsrworkflow: Action 'Engineering/ClientProtection/Backup's
log will be in '/nsr/logs/policy/Engineering/ClientProtection/Backup_032028.raw'.
123325:nsrworkflow: Action 'Engineering/ClientProtection/Backup'
succeeded.
133553:nsrworkflow: Workflow 'Engineering/ClientProtection'
succeeded.
```

API request

The same on-demand backup is performed in the following example by using the NetWorker REST API:

```
API request:
POST /nwrestapi/v2/global/protectionpolicies/Engineering/workflows/ClientProtection/op/backup
{
  "clients": ["saturn:/etc/hosts"]
}
```

API response

The following response is received from the API:

```
201 Created
```

Retrieving information about the backup job

The location header in the response from the above API will contain a URL to the job that was created. Use the following GET request to that URL to retrieve information about the job:

```
GET /nwrestapi/v2/global/jobs/64081
```

The following information is returned:

```
{
  "adhocJob": false,
  "command": "/usr/sbin/nsrworkflow -p Engineering -w
ClientProtection -L -c saturn:/etc/hosts",
  "dependentJobIds": [
    0
  ],
  "id": 64081,
  "itemIdLong": 64081,
  "links": [
    {
      "href": "https://mars:9090/nwrestapi/v2/global/jobs/64081/op/
cancel",
      "title": "Cancel job"
    }
  ],
  "logFile": "/nsr/logs/policy/Engineering/
workflow_ClientProtection_064081.raw",
  "name": "Engineering",
  "ndmp": false,
  "parentJobId": 0,
  "previousJobId": 0,
  "rootParentJobId": 0,
  "runOnHost": "mars",
  "siblingJobIds": [],
  "startTime": "2016-05-17T12:34:07-04:00",
  "state": "Active",
  "stopped": false,
  "tenant": "",
  "type": "workflow job"
}
```

Viewing all backups for a client

This section demonstrates how to view all backups for a client by using the NetWorker REST API.

Command line example

The following example command uses the `mminfo` command to view the backups for a client:

```
# mminfo -avot -q "client=saturn"
volume  type  client  date      time      size  ssid
fl      lvl  name
mars.001  adv_file mars    05/17/16 11:41:38 6 KB 4030413746 cb
manual /etc/hosts
...
```

API request

The same request is created in the following example by using the NetWorker REST API:

```
GET /nwrestapi/v2/global/clients/
82.0.31.22.0.0.0.0.33.12.41.87.128.222.109.201/backups
```

Note

In the command line example, the name of the client is used. However when using the REST API, the resourceId of the client must be used instead, as show in this example URL. This can be retrieved from the resourceId element in the client resource, as shown in the API response section of [Viewing all clients](#) on page 18.

API response

The following response is received from the API:

```
{
  "backups": [
    {
      "attributes": [
        {
          "key": "*backup start time",
          "values": [
            "1463499698"
          ]
        },
        {
          "key": "*ss clone retention",
          "values": [
            "1463499698: 1463499698: -2"
          ]
        },
        {
          "key": "save set features",
          "values": [
            "CLIENT_SAVETIME"
          ]
        }
      ],
      "browseTime": "2038-01-18T22:14:06-05:00",
      "clientHostname": "mars",
      "clientId": "b7d73be0-00000004-57290c2f-57290c2e-00010c00-80c64a29",
      "completionTime": "2016-05-17T11:41:38-04:00",
      "creationTime": "2016-05-17T11:41:38-04:00",
      "fileCount": 3,
      "id": "67f3852c-00000006-f03b3bb2-573b3bb2-00120c00-80c64a29",
      "instances": [
        {
          "clone": false,
          "id": "1463499698",
          "status": "Browsable",
          "volumeIds": [
            "4264957582"
          ]
        }
      ],
      "level": "Manual",
      "links": [
        {
          "href": "https://mars:9090/nwrestapi/v2/global/clients/82.0.31.22.0.0.0.33.12.41.87.128.222.109.201/backups/67f3852c-00000006-f03b3bb2-573b3bb2-00120c00-80c64a29",
          "rel": "item"
        }
      ],
      "name": "\\etc\\hosts",
      "retentionTime": "2038-01-18T22:14:06-05:00",
      "saveTime": "2016-05-17T11:41:38-04:00",
      "shortId": "4030413746",
      "size": {
```

```

        "unit": "Byte",
        "value": 6736
      },
      "type": "File"
    }
    ...
  ]
}

```

Working with VMware-based clients

Because of limited command line support for VMware client operations, the examples in this section use NetWorker Management Console (NMC) examples, rather than command line examples.

Note

The NetWorker REST API does not have an ability to register a vCenter or a vProxy object, so we assume that these objects have been already configured in NMC.

Creating a protection group for VMware-based clients

This section demonstrates how to create a protection group for VMware-based clients by using the NetWorker REST API.

NMC example

To create a protection group in NMC:

1. Navigate to the **Protection** tab.
2. Select **Groups**.
3. Right-click and select **New** and enter a name for the group.
4. Set the group type to **VMware** and sub-type to **All**.
5. Select the vCenter to be used.
6. Browse the vCenter to select individual VMs and containers such as resource pools.

API request

The same protection group is created in the following example by using the NetWorker REST API:

```

POST /nwrestapi/v2/global/protectiongroups
{
  "name": "ApiGroup",
  "vmwareWorkItemSelection": {
    "containerMorefs": [
      "resgroup-7675"
    ],
    "vCenterHostname": "hostname.company.com",
    "vmUuids": [
      "50085479-6e0f-2383-7cb8-15883df49fde"
    ],
    "vmdks": []
  },
  "workItemQueries": [],
  "workItemSource": "Static",
  "workItemSubType": "All",
}

```

```
"workItemType": "VMware"
}
```

This protection group contains two selection items:

- A VM with a specific UUID—50085479-6e0f-2383-7cb8-15883df49fde. Resources such as `/vmware/vcenters/{vcenter-hostname}/vms`, which returns a list of VMs on the specified vCenter, can be used to retrieve the UUID for a VM.
- A container with a specified Managed Object Reference (MoRef)—`resgroup-7675`
When a container is included, then all VMs included in this container will be backed up, even if they are added after the protection group was created. The NetWorker REST API does not expose all the containers present in the vCenter. These should be obtained through another mechanism such as directly from the vCenter.

API response

The following response is received from the API:

```
201 Created
```

Creating a protection policy for VMware-based clients

This section demonstrates how to create a protection policy for VMware-based clients by using the NetWorker REST API.

NMC example

To create a protection group in NMC:

1. Navigate to the **Protection** tab.
2. Select **Policies**.
3. Right-click and select **New** and enter a name for the policy.
4. Select **Create a new workflow** and enter a name for the workflow.
5. Associate the new workflow with a protection group. In this example, we will associate it with the ApiGroup created in [Creating a protection group for VMware-based clients](#) on page 29.
6. Create an action, enter a name for the action, and set the action type to **Backup** and the backup subtype to **VMware (vProxy)**.

API request

The same protection policy is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/protectionpolicies
{
  "name": "ApiPolicy",
  "workflows": [
    {
      "actions": [
        {
          "actionSpecificData": {
            "backup": {
              "backupSpecificData": {
                "vmwareVProxy": {
                  "quiesceApp": false,
```

```

        "destinationPool": "Data Domain Default"
    },
    "destinationStorageNodes": [
        "nsrserverhost"
    ]
},
"name": "ApiAction",
"scheduleActivities": [
    "incr",
    "incr",
    "incr",
    "incr",
    "incr",
    "incr"
],
"schedulePeriod": "Week"
},
],
"name": "ApiWorkflow",
"protectionGroups": [
    "ApiGroup"
]
}
]
}

```

Some properties, such as `scheduleActivities`, are optional and will default to the same values as those shown in NMC.

After the policy resource is created, you can perform a GET for the resource URL returned in the location header to see the full resource with all populated default values.

API response

The following response is received from the API:

```
201 Created
```

Adding a new item to a for VMware-based protection group

This section demonstrates how to add a new item, such as a new VM, to a VMware-based protection group by using the NetWorker REST API.

NMC example

To add a new item to a VMware-based protection group in NMC:

1. Navigate to the **Protection** tab.
2. Select **Groups**.
3. Select the group.
4. Right-click and select **Properties** and add the item.

API request

To add an item to a VMware-based protection group by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/protectiongroups/ApiGroup/op/
updatevmwareworkitems
```

```
{
  "addWorkItems": {
    "vCenterHostname": "hostname.company.com",
    "vmUuids": [
      "5008edba-edfe-127c-1d00-68830b14b0fc"
    ]
  }
}
```

Resources such as `/vmware/vcenters/{vcenter-hostname}/vms`, which returns a list of VMs on the specified vCenter, can be used to retrieve the UUID for a VM.

API response

The following response is received from the API:

```
204 No Content
```

Performing a VMware-based on-demand backup with an override

This section demonstrates how to perform a VMware-based on-demand backup with an override by using the NetWorker REST API.

NMC example

With NMC, you can run a backup only for a specific VM, container, or subset of VMs in a container:

1. Navigate to the **Protection** tab.
2. Double-click **VMware View**.
3. Select a VM or container associated with a protection group.
4. Right-click and select **Start Individual Client....**
5. The **Start Workflow** dialog will appear, where you can verify or adjust the selection.

API request

The same on-demand backup is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/protectionpolicies/ApiPolicy/workflows/
ApiWorkflow/op/backup
{
  "vmwareWorkItemSelection" : {
    "vCenterHostname" : "hostname.company.com",
    "vmUuids" : ["50085479-6e0f-2383-7cb8-15883df49fde"]
  }
}
```

API response

The following response is received from the API:

```
201 Created
```

The API response will contain a Job URI in the location header. This URI can be used to monitor the job by using `GET jobs/{jobId}`, and to cancel it if necessary by using `POST /jobs/{jobId}/op/cancel`.

Performing an on-demand backup for a specific VM

This section demonstrates how to perform an on-demand backup for a specific VM by using the NetWorker REST API.

NMC example

To perform an on-demand backup for a specific VM in NMC:

1. Navigate to the **Protection** tab.
2. Double-click **VMware View**.
3. Select a VM or container associated with a protection group.
4. Right-click and select **Start Individual Client...**
5. The **Start Workflow** dialog will appear, where you can select a policy or workflow if the VM is protected by more than one.

API request

The same on-demand backup is created in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/vmware/vcenters/hostname.company.com/vms/
5008edba-edfe-127c-1d00-68830b14b0fc/op/backup
{
  "policy" : "ApiPolicy",
  "workflow" : "ApiWorkflow"
}
```

API response

The following response is received from the API:

```
201 Created
```

The API response will contain a Job URI in the location header. This URI can be used to monitor the job by using `GET jobs/{jobId}`, and to cancel it if necessary by using `POST /jobs/{jobId}/op/cancel`.

Recovering a VM to a previous state

This section demonstrates how to recover a VM to a previous state (revert) by using the NetWorker REST API.

NMC example

To recover a VM to a previous state (revert) in NMC:

1. Navigate to the **Recover** tab.
2. Right-click and select **New recover** and enter a name for the group.
3. In the **Recover** wizard, select **Virtual Machine Recovery**.
4. Select a vCenter.
5. Select the VM and backup to be recovered.
6. For **Virtual Machine Recovery Type**, select **Revert a Virtual Machine**.
This example assumes that all disks will be recovered.

API request

The same recovery is performed in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/vmware/vcenters/hostname.company.com/protectedvms/50085479-6e0f-2383-7cb8-15883df49fde/backups/9524aeb0-00000006-f824e4c1-5824e4c1-00090c00-80c64a29/instances/1478812865/op/rrecover
{
  "recoverMode" : "Revert",
  "powerOn": true,
  "reconnectNic": false
}
```

When making this API request, start with a backup (saveset) of the VM. This can be retrieved using `GET /backups/{backupId}/instances`.

API response

The following response is received from the API:

```
201 Created
```

The API response will contain a Job URI in the location header. This URI can be used to monitor the job by using `GET jobs/{jobId}`, and to cancel it if necessary by using `POST /jobs/{jobId}/op/cancel`.

Recovering a VM as a new VM

This section demonstrates how to recover a VM as a new VM by using the NetWorker REST API.

NMC example

To recover a VM as a new VM in NMC:

1. Navigate to the **Recover** tab.
2. Right-click and select **New recover** and enter a name for the group.
3. In the **Recover** wizard, select **Virtual Machine Recovery**.
4. Select a vCenter.
5. Select the VM and backup to be recovered.
6. For **Virtual Machine Recovery Type**, select **Virtual Machine Recovery**.
7. Select the location where the VM should be recovered.
8. Select the disks to be recovered.

API request

The same recovery is performed in the following example by using the NetWorker REST API:

```
POST /nwrestapi/v2/global/vmware/vcenters/hostname.company.com/protectedvms/50085479-6e0f-2383-7cb8-15883df49fde/backups/9524aeb0-00000006-f824e4c1-5824e4c1-00090c00-80c64a29/op/recover
{
  "computeResourceMoref" : "domain-s7",
  "datacenterMoref": "datacenter-2",
  "datastoreMoref" : "datastore-11",
  "disks" : [
```

```

{
  "datastoreMoref": "datastore-11",
  "key": "2000",
  "name": "Hard disk 1"
},
{
  "datastoreMoref": "datastore-11",
  "key": "2001",
  "name": "Hard disk 2"
}
],
"hostMoref": "host-9",
"jobName": "jobName",
"powerOn": false,
"reconnectNic": false,
"recoverMode": "New",
"vCenterHostname": "hostname.company.com",
"vmName": "newVmName"
}

```

This example requires information about the VMware recovery location and disk configuration, which can only be obtained from the vCenter outside of the API. If only recovery to the original VM location is required, most of the parameters in this example, including the disks, can be retrieved from the VM state recorded during the backup. This information is available in the `vmInformation` property of the backup object.

API response

The following response is received from the API:

```
201 Created
```

The API response will contain a Job URI in the location header. This URI can be used to monitor the job by using `GET jobs/{jobId}`, and to cancel it if necessary by using `POST /jobs/{jobId}/op/cancel`.

APPENDIX A

Troubleshooting

This appendix includes the following topics:

- [NetWorker REST API log files](#).....38
- [NetWorker REST API log file management](#)..... 38

NetWorker REST API log files

The following table provides a summary of the log files available for the NetWorker REST API.

Component	File name and default location	Description
Installation log Tomcat Access log Apache Catalina log	Refer to the NetWorker Authentication Service logs for these log files.	The NetWorker REST API service is deployed in the same Apache Tomcat container as the NetWorker authentication service. The NetWorker REST API uses the same installation log, Tomcat Access log and Apache Catalina log with NetWorker Authentication service.
NetWorker REST API log	- Linux: <code>/nsr/logs/restapi/restapi.log</code> - Windows: <code>C:\Program Files\EMC NetWorker\nsr\logs\restapi\restapi.log</code>	Main NetWorker REST API log file.

NetWorker REST API log file management

The NetWorker REST API uses the Logback API to manage log files.

To modify how NetWorker REST API manages the `restapi.log` log file, edit the `logback.xml` file, which is found in the following locations:

- Linux: `/nsr/authc/webapps/nwrestapi/WEB-INF/classes`
- Windows: `C:\Program Files\EMC NetWorker\nsr\authc-server\tomcat\webapps\nwrestapi\WEB-INF\classes`

This section describes how to modify the commonly used log attributes in the `logback.xml` file. Logback project documentation provides more detailed information about each attribute in the `logback.xml` file.

Modifying the logging level

In the `logger` configuration element, the value of the `level` attribute defines the level of logging that the NetWorker REST API writes to the log files. By default, the NetWorker REST API sets the logging level to `info` and messages appear in the log files. There are five standard log levels: `trace`, `debug`, `info`, `warn`, `error`.

To change the logging level to `error`, modify the `level` attribute to appear as follows:

```
<logger name="com.emc.nw.webapi" level="error"/>
```

Modifying the rollover period

In the `rollingPolicy` configuration element, the `fileNamePattern` element can be used to define the rollover period of the `restapi.log` file. When the log file reaches the end of a rolling period, the NetWorker REST API renames the log file for archival purposes and creates new log file. The rollover period is inferred from the value of `fileNamePattern`. By default, the NetWorker REST API sets the rollover period to `monthly`.

To change the rollover period to daily, modify the `fileNamePattern` element to appear as follows:

```
<fileNamePattern>${logdir}/restapi_%d.log</fileNamePattern>
```

Please refer to Logback project document on how to modify the rollover period.

Modifying the number of rollover log files

In the `rollingPolicy` configuration element, the `maxHistory` element defines the number of `restapi.log` rollover log files that the NetWorker REST API maintains. It controls the maximum number of archive files to keep, deleting older files. By default, the NetWorker REST API maintains six rollover log files.

